Ex NO: 1 Date:

Study of Various Network Commands used in Linux and Windows

Basic Commands (windows)

1. arp -a :
   Displays the ARP table, showing IP addresses and corresponding MAC addresses

2. hostname :
   Displays the computer's hostname

3. ipconfig/all (windows)
   shows complete TCP/IP configuration including IP, gateway, DNS, DHP and adapter details.

4. nbtstat -a <hostname>
   Displays NetBIOS information for a remote computer.

5. netstat
   Displays active network connections and network statistics

6. netstat -r
   Displays the routing table

7. nslookup <Domain>
   Queries DNS to find the IP address or DNS records of a Domain

8. pathping <host>
   combines ping and tracert to trace the route and analyze packet loss.

Outputs:

1)

```
C:\Users\REC>arp -a

Interface: 172.16.8.56 --- 0x9
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.8.7            d4-3d-7e-ad-3e-bf     dynamic
  172.16.8.48           f8-bc-12-69-80-01     dynamic
  172.16.8.62           f8-bc-12-8d-ed-be     dynamic
  172.16.10.66          d8-bb-c1-c5-6a-9d     dynamic
  172.16.10.84          d8-bb-c1-c5-c8-99     dynamic
  172.16.11.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  230.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

2)

```
C:\Users\REC>hostname
DESKTOP-ECC1JEK
```

3)

```
C:\Users\REC>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d988:bcf0:4035:27e8%9
   IPv4 Address. . . . . . . . . . . : 172.16.8.56
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

4)

```
C:\Users\REC>nbtstat -r

    NetBIOS Names Resolution and Registration Statistics
    ----------------------------------------------------

    Resolved By Broadcast     = 0
    Resolved By Name Server   = 0

    Registered By Broadcast   = 54
    Registered By Name Server = 0
```

5)

```
C:\Users\REC>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.8.56:63464      sf-in-f188:5228        ESTABLISHED
  TCP    172.16.8.56:63602      104.18.32.47:https     ESTABLISHED
  TCP    172.16.8.56:63627      a23-44-10-56:https     CLOSE_WAIT
  TCP    172.16.8.56:63637      hkg12s32-in-f3:https   ESTABLISHED
  TCP    172.16.8.56:63639      ym-in-f94:https        ESTABLISHED
  TCP    172.16.8.56:63640      4.213.25.240:https     ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:1521  DESKTOP-ECC1JEK:49674  ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:49674  DESKTOP-ECC1JEK:1521  ESTABLISHED
```

9 Ping <host>

Tests connectivity to another host using ICMP Echo Request packets.

10 route

Displays or modifies the IP routing table.

6)

```
>netstat -r
===========================================================================
Interface List
  9...00 27 0e 13 eb f8 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.8.1    172.16.8.56     291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1     331
       172.16.8.0    255.255.252.0         On-link     172.16.8.56     291
      172.16.8.56  255.255.255.255         On-link     172.16.8.56     291
  172.16.11.255  255.255.255.255         On-link     172.16.8.56     291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link     172.16.8.56     291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link     172.16.8.56     291
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.8.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  9    291 fe80::/64                On-link
  9    291 fe80::d988:bcf0:4035:27e8/128
                                    On-link
  1    331 ff00::/8                 On-link
  9    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

7)

```
C:\Users\REC>nslookup google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:808::200e
          142.250.66.14
```

8)

```
C:\Users\REC>pathping google.com

Tracing route to google.com [172.217.24.14]
over a maximum of 30 hops:
  0  DESKTOP-ECC1JEK [172.16.8.56]
  1  172.16.8.1
  2  137.97.163.129
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-ECC1JEK [172.16.8.56]
                                0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%   0/ 100 =  0%  172.16.8.1
                                0/ 100 =  0%   |
  2    2ms     0/ 100 =  0%   0/ 100 =  0%  137.97.163.129

Trace complete.
```

9)

```
C:\Users\REC>ping google.com

Pinging google.com [142.250.205.142] with 32 bytes of data:
Reply from 142.250.205.142: bytes=32 time=4ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118
Reply from 142.250.205.142: bytes=32 time=2ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118

Ping statistics for 142.250.205.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

10)

```
>route print
===========================================================================
Interface List
  9...00 27 0e 13 eb f8 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.8.1    172.16.8.56     291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1     331
       172.16.8.0    255.255.252.0         On-link     172.16.8.56     291
      172.16.8.56  255.255.255.255         On-link     172.16.8.56     291
  172.16.11.255  255.255.255.255         On-link     172.16.8.56     291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link     172.16.8.56     291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link     172.16.8.56     291
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.8.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  9    291 fe80::/64                On-link
  9    291 fe80::d988:bcf0:4035:27e8/128
                                    On-link
  1    331 ff00::/8                 On-link
  9    291 ff00::/8                 On-link
```

Linux Commands

1) ip command

Displays and manages IP addresses, network interfaces, and routing

a) ip address show

Displays all network interfaces and their IP addresses.

b) Assign an Ip address:

ip address add 192.168.1.254/24 dev enp0s3

assigns an IP address to the interface

c) Remove an IP Address:

ip address del 192.168.1.254/24 dev enp0s3

Deletes the assigned IP address

d) ip link set enp0s3 up

Enables the network interface

e) ip link set enp0s3 down

Disables the network interface

f) ip link set enp0s3 promisc on

Allows the interface to capture all network traffic

g) ip route add default via 10.0.2.2 dev enp0s3

Adds the default gateway

h) ip route add 192.168.1.0/24 via 10.0.2.2 dev enp0s3

Adds a route to a specific network

i) ip rate del 192.168.1.0/24 dev enp0s3

Removes the route

a)

```
[root@localhost ~]# ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85830sec preferred_lft 85830sec
    inet6 fe80::a00:27ff:fec1:3ab4/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

b)

```
[root@localhost ~]# ip address add 192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85400sec preferred_lft 85400sec
    inet 192.168.1.254/24 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec1:3ab4/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

c)

```
[root@localhost ~]# ip address del  192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85322sec preferred_lft 85322sec
    inet6 fe80::a00:27ff:fec1:3ab4/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

d)

```
[root@localhost ~]# ip link set enp0s3 up
[root@localhost ~]# ip link show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
[root@localhost ~]#
```

e)

```
[root@localhost ~]# ip link set enp0s3 down
[root@localhost ~]# ip link show enp0s3
2: enp0s3: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT
 group default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
[root@localhost ~]#
```

f)

```
[root@localhost ~]# ip link set enp0s3 promisc on
[root@localhost ~]# ip link show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,PROMISC> mtu 1500 qdisc fq_codel state DOWN mode
 DEFAULT group default qlen 1000
    link/ether 08:00:27:c1:3a:b4 brd ff:ff:ff:ff:ff:ff
[root@localhost ~]#
```

g)

```
[root@localhost ~]# ip route add default via 10.0.2.2 dev enp0s3
[root@localhost ~]# ip route
default via 10.0.2.2 dev enp0s3
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[root@localhost ~]#
```

i) ip route get 10.10.1.4

Shows which route packets will take

2) ifconfig :

Displays and configures network interfaces.

3) mtr

shows the route packets take and network delays.

i) mtr -g google.com

Displays the network route to the destination using numeric IP addresses instead of hostnames.

ii) mtr -b google.com

Displays the network route showing both hostnames and their corresponding IP addresses for each hop.

ii) mtr -c 10 google.com

Sends 10 probe packets to the destination and displays statistics such as packet loss and response time for each network hop

4) tcpdump

Captures and analyzes network packets on a selected interface

i) tcpdump -D

lists all available network interfaces for packet capture

h)

```
[root@localhost ~]# ip route add 192.168.1.0/24 via 10.0.2.2 dev enp0s3
[root@localhost ~]# ip route
default via 10.0.2.2 dev enp0s3
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.1.0/24 via 10.0.2.2 dev enp0s3
[root@localhost ~]#
```

i)

```
[root@localhost ~]# ip route del 192.168.1.0/24 dev enp0s3
[root@localhost ~]# ip route
default via 10.0.2.2 dev enp0s3
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[root@localhost ~]#
```

j)

```
[root@localhost ~]# ip route get 10.10.1.4
10.10.1.4 via 10.0.2.2 dev enp0s3 src 10.0.2.15 uid 0
    cache
[root@localhost ~]#
```

2)

```
root@localhost ~]# ifconfig
np0s3: flags=4419<UP,BROADCAST,RUNNING,PROMISC,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fec1:3ab4  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:c1:3a:b4  txqueuelen 1000  (Ethernet)
        RX packets 176  bytes 15760 (15.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 331  bytes 31883 (31.1 KiB)
        TX errors 0  dropped 2 overruns 0  carrier 0  collisions 0

o: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

3i)

```
                    My traceroute  [v0.94]
localhost.localdomain (10.0.2.15) -> google.com        2026-07-22T20:44:12+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                  Packets                Pings
 Host                          Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. _gateway                    0.0%    76    0.7   1.9   0.5   7.5   1.4
 2. (waiting for reply)
 3. (waiting for reply)
 4. (waiting for reply)
 5. (waiting for reply)
 6. (waiting for reply)
 7. (waiting for reply)
 8. (waiting for reply)
 9. (waiting for reply)
10. (waiting for reply)
11. (waiting for reply)
12. (waiting for reply)
13. pnmaaa-aw-in-f14.1e100.net  14.9%    75   61.8 179.5  19.8 1447. 229.9
```

ii)

```
                    My traceroute  [v0.94]
localhost.localdomain (10.0.2.15) -> google.com        2026-07-22T20:47:02+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                  Packets                Pings
 Host                          Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. _gateway (10.0.2.2)         0.0%    40    1.2   1.6   0.6   2.6   0.4
 2. (waiting for reply)
```

iii)

```
                    My traceroute  [v0.94]
localhost.localdomain (10.0.2.15) -> google.com        2026-07-22T20:48:24+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                  Packets                Pings
 Host                          Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. _gateway                    0.0%    10    2.3   1.6   0.9   2.3   0.4
 2. (waiting for reply)
```

ii) tcpdump -i enp0s3

Captures packets on the enp0s3 interface

iii) tcpdump -i enp0s3 -c 10

captures only the first 10 packets on enp0s3

iv) tcpdump -i enp0s3 host 8.8.8.8

Captures traffic to and from the specified host

v) tcpdump -i enp0s3 src host 8.8.8.8

Captures only incoming traffic from the specified host.

vi) tcpdump -i enp0s3 dst host 8.8.8.8

Captures only outgoing traffic to the specified network.

vii) tcpdump -i enp0s3 port 53

Captures traffic on DNS port 53

viii) tcpdump -i enp0s3 net 10.1.0.0/24

Captures traffic belonging to the specified network.

ix) tcpdump -i enp0s3 host 8.8.8.8 and port 53

Captures DNS traffic to or from the specified host.

```
root@localhost:~# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
```

```
0.48.24.10.in-addr.arpa. (42)
:23:24.794429 IP 10.24.48.39.domain > localhost.localdomain.59949: 26305 NXDom
n 0/0/0 (42)
:23:25.127614 IP localhost.localdomain.35873 > ec2-13-200-20-166.ap-south-1.co
ute.amazonaws.com.ntp: NTPv4, Client, length 48
:23:25.142033 IP localhost.localdomain.45229 > 10.24.48.39.domain: 11678+ PTR?
66.20.200.13.in-addr.arpa. (44)
:23:25.145734 IP 10.24.48.39.domain > localhost.localdomain.45229: 11678 1/0/0
TR ec2-13-200-20-166.ap-south-1.compute.amazonaws.com. (108)
:23:25.201090 IP ec2-13-200-20-166.ap-south-1.compute.amazonaws.com.ntp > loca
st.localdomain.35873: NTPv4, Server, length 48
:23:25.505969 IP localhost.localdomain.57711 > ec2-13-126-27-131.ap-south-1.co
ute.amazonaws.com.ntp: NTPv4, Client, length 48
:23:25.558918 IP localhost.localdomain.33295 > 10.24.48.39.domain: 12020+ PTR?
31.27.126.13.in-addr.arpa. (44)
:23:25.562744 IP 10.24.48.39.domain > localhost.localdomain.33295: 12020 1/0/0
TR ec2-13-126-27-131.ap-south-1.compute.amazonaws.com. (108)
23:25.571549 IP ec2-13-126-27-131.ap-south-1.compute.amazonaws.com.ntp > loca
st.localdomain.57711: NTPv4, Server, length 48

packets captured
packets received by filter
ackets dropped by kernel
```

```
1:22:18.555865 IP localhost.localdomain.37586 > ec2-15-207-248-194.ap-south-1.c
mpute.amazonaws.com.ntp: NTPv4, Client, length 48
1:22:18.592770 IP localhost.localdomain.46164 > 10.24.48.39.domain: 21981+ PTR?
194.248.207.15.in-addr.arpa. (45)
1:22:18.599483 IP 10.24.48.39.domain > localhost.localdomain.46164: 21981 1/0/0
PTR ec2-15-207-248-194.ap-south-1.compute.amazonaws.com. (110)
1:22:18.600232 IP localhost.localdomain.35485 > 10.24.48.39.domain: 2824+ PTR?
.2.0.10.in-addr.arpa. (40)
1:22:18.609889 IP 10.24.48.39.domain > localhost.localdomain.35485: 2824 NXDoma
0/0/0 (40)
1:22:18.628361 IP ec2-15-207-248-194.ap-south-1.compute.amazonaws.com.ntp > loc
lhost.localdomain.37586: NTPv4, Server, length 48
1:22:18.696415 IP localhost.localdomain.51304 > 10.24.48.39.domain: 7983+ PTR?
.48.24.10.in-addr.arpa. (42)
1:22:18.702339 IP 10.24.48.39.domain > localhost.localdomain.51304: 7983 NXDoma
0/0/0 (42)
1:22:18.702340 IP 10.24.48.39.domain > localhost.localdomain.51304: 7983 NXDoma
0/0/0 (42)
1:22:20.826875 IP localhost.localdomain.41163 > ec2-13-126-27-131.ap-south-1.co
pute.amazonaws.com.ntp: NTPv4, Client, length 48
packets captured
packets received by filter
packets dropped by kernel
```

```
seq 4, length 64
11:51:09.978732 IP dns.google > localhost.localdomain: ICMP echo reply, id 6, se
q 4, length 64
11:51:12.103861 ARP, Request who-has _gateway tell localhost.localdomain, length
 28
11:51:12.104518 ARP, Reply _gateway is-at 52:54:00:12:35:02 (oui Unknown), lengt
h 46
11:51:12.196071 IP localhost.localdomain.58873 > 10.24.48.39.domain: 20022+ PTR?
 2.2.0.10.in-addr.arpa. (39)
11:51:12.208608 IP 10.24.48.39.domain > localhost.localdomain.58873: 20022 NXDom
ain 0/0/0 (39)
11:51:25.694712 IP localhost.localdomain.45785 > ec2-13-200-20-166.ap-south-1.co
mpute.amazonaws.com.ntp: NTPv4, Client, length 48
11:51:25.745120 IP localhost.localdomain.47838 > 10.24.48.39.domain: 16798+ PTR?
 166.20.200.13.in-addr.arpa. (44)
11:51:25.747237 IP ec2-13-200-20-166.ap-south-1.compute.amazonaws.com.ntp > loca
lhost.localdomain.45785: NTPv4, Server, length 48
11:51:25.808736 IP 10.24.48.39.domain > localhost.localdomain.47838: 16798 1/0/0
 PTR ec2-13-200-20-166.ap-south-1.compute.amazonaws.com. (108)
^C
22 packets captured
22 packets received by filter
0 packets dropped by kernel
```

```
istening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:31:47.134270 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 1, length 64
11:31:48.139185 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 2, length 64
11:31:49.138129 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 3, length 64
11:31:50.134770 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 4, length 64
11:31:51.180571 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 5, length 64
11:31:52.158959 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 6, length 64
11:31:53.216345 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 7, length 64
11:31:54.189844 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 8, length 64
11:31:55.216202 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 9, length 64
11:31:56.214919 IP dns.google > localhost.localdomain: ICMP echo reply, id 3, se
q 10, length 64
```

```
q 3, length 64
11:29:10.547865 IP localhost.localdomain > dns.google: ICMP echo request, id 2,
seq 4, length 64
11:29:10.613945 IP dns.google > localhost.localdomain: ICMP echo reply, id 2, se
q 4, length 64
11:29:11.549982 IP localhost.localdomain > dns.google: ICMP echo request, id 2,
seq 5, length 64
11:29:11.615772 IP dns.google > localhost.localdomain: ICMP echo reply, id 2, se
q 5, length 64
10 packets captured
10 packets received by filter
0 packets dropped by kernel
```

x) tcpdump -i enp0s3 host and port 443

Captures HTTPS traffic from the specified host.

xi) tcpdump -i enp0s3 port not 53 and not 25

Captures all traffic except DNS and SMTP traffic

ping:

i) ping ~~google~~.8.8.8.8

Continuously checks connectivity to the specified host

ii) ping -c 10 8.8.8.8

Sends exactly 10 ICMP Echo Requests to the specified host.

Configuring an Ethernet Connection using nmcli

1. nmcli connection show

Displays all available NetworkManager connection profiles and their associated devices.

2. nmcli connection add con-name <connection-name> ifname <device-name> type ethernet

Creates a new Ethernet connection profile for the specified network interface

```
packets captured
packets received by filter
packets dropped by kernel
```

```
138.177.178.192.in-addr.arpa. (46)
11:54:19.533861 IP 10.24.48.39.domain > localhost.localdomain.49898: 49190 1/0/0
PTR lcdelw-in-f138.1e100.net. (84)
11:54:20.529116 IP localhost.localdomain.37815 > 10.24.48.39.domain: 49367+ PTR?
138.177.178.192.in-addr.arpa. (46)
11:54:20.534871 IP 10.24.48.39.domain > localhost.localdomain.37815: 49367 1/0/0
PTR lcdelw-in-f138.1e100.net. (84)
^C
14 packets captured
14 packets received by filter
0 packets dropped by kernel
```

```
2:00:07.265548 IP 10.24.48.39.domain > localhost.localdomain.43648: 41815 3/0/0
PTR nrt20s02-in-f14.1e100.net., PTR nrt20s02-in-f14.1e100.net., PTR tzdelb-ay-i
-f14.1e100.net. (128)
2:00:08.390163 IP localhost.localdomain.42963 > 10.24.48.39.domain: 51821+ PTR?
14.26.217.172.in-addr.arpa. (44)
2:00:08.395516 IP 10.24.48.39.domain > localhost.localdomain.42963: 51821 3/0/0
PTR nrt20s02-in-f14.1e100.net., PTR nrt20s02-in-f14.1e100.net., PTR tzdelb-ay-i
-f14.1e100.net. (162)
^C
2 packets captured
2 packets received by filter
packets dropped by kernel
```

```
12:07:25.745409 IP 142.251.151.119.https > localhost.localdomain.50324: Flags [P
.], seq 1281:2818, ack 518, win 65535, length 1537
12:07:25.745544 IP localhost.localdomain.50324 > 142.251.151.119.https: Flags [.
], ack 2818, win 65023, length 0
12:07:25.747257 IP localhost.localdomain.50324 > 142.251.151.119.https: Flags [P
.], seq 518:598, ack 2818, win 65023, length 80
10 packets captured
24 packets received by filter
0 packets dropped by kernel
```

```
12:09:30.691111 IP localhost.localdomain > dns.google: ICMP echo request, id 14,
 seq 5, length 64
12:09:30.735348 IP dns.google > localhost.localdomain: ICMP echo reply, id 14, s
eq 5, length 64
12:09:33.928615 ARP, Request who-has _gateway tell localhost.localdomain, length
 28
12:09:33.929471 ARP, Reply _gateway is-at 52:54:00:12:35:02 (oui Unknown), lengt
h 46
^C
16 packets captured
16 packets received by filter
0 packets dropped by kernel
```

```
[root@localhost ~]# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
From 192.168.1.100 icmp_seq=1 Destination Host Unreachable
From 192.168.1.100 icmp_seq=2 Destination Host Unreachable
From 192.168.1.100 icmp_seq=3 Destination Host Unreachable
From 192.168.1.100 icmp_seq=4 Destination Host Unreachable
From 192.168.1.100 icmp_seq=5 Destination Host Unreachable
From 192.168.1.100 icmp_seq=6 Destination Host Unreachable
```

```
[root@localhost ~]# ping -c 10 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
From 192.168.1.100 icmp_seq=1 Destination Host Unreachable
From 192.168.1.100 icmp_seq=2 Destination Host Unreachable
From 192.168.1.100 icmp_seq=3 Destination Host Unreachable
From 192.168.1.100 icmp_seq=4 Destination Host Unreachable
From 192.168.1.100 icmp_seq=5 Destination Host Unreachable
From 192.168.1.100 icmp_seq=6 Destination Host Unreachable
From 192.168.1.100 icmp_seq=7 Destination Host Unreachable
From 192.168.1.100 icmp_seq=8 Destination Host Unreachable
From 192.168.1.100 icmp_seq=9 Destination Host Unreachable
From 192.168.1.100 icmp_seq=10 Destination Host Unreachable

--- 8.8.8.8 ping statistics ---
10 packets transmitted, 0 received, +10 errors, 100% packet loss, time 9196ms
pipe 4
```

```
[root@localhost ~]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
enp0s3  1cdf9c34-f850-3a2-ae1b-c6971bd22b92  ethernet  enp0s3
```

```
[root@localhost ~]# nmcli connection add con-name Internal-LAN ifname enp0s3 typ
e ethernet
Connection 'Internal-LAN' (9d2bb058-c121-4175-8bd3-03d422fad409) successfully ad
ded.
```

```
[root@localhost ~]# nmcli connection modify "wired connection 1" connection.id
internal-LAN2
[root@localhost ~]# nmcli connection show
NAME           UUID                                  TYPE      DEVICE
enp0s3         1cdf9c34-f850-3a2-ae1b-c6971bd22b92  ethernet  enp0s3

Internal-LAN   db5678ae-70df-4faa-8935-727d7745ef82  ethernet  --
Internal-LAN2  9d2bb058-c121-4175-8bd3-03d422fad409  ethernet  --
```

3. nmcli connection modify "wired Connection 1" connection.id Internal-LAN2

modifies the settings of an existing network connection profile

4. nmcli connection modify "Internal-LAN" ipv4.method auto

Configures the connection to automatically obtain an IPV4 address using

5. nmcli connection modify "Internal-LAN ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200

Assigns a static IPV4 address, gateway DNS server, and DNS Search Domain to the connection

6. nmcli connection up Internal-LAN

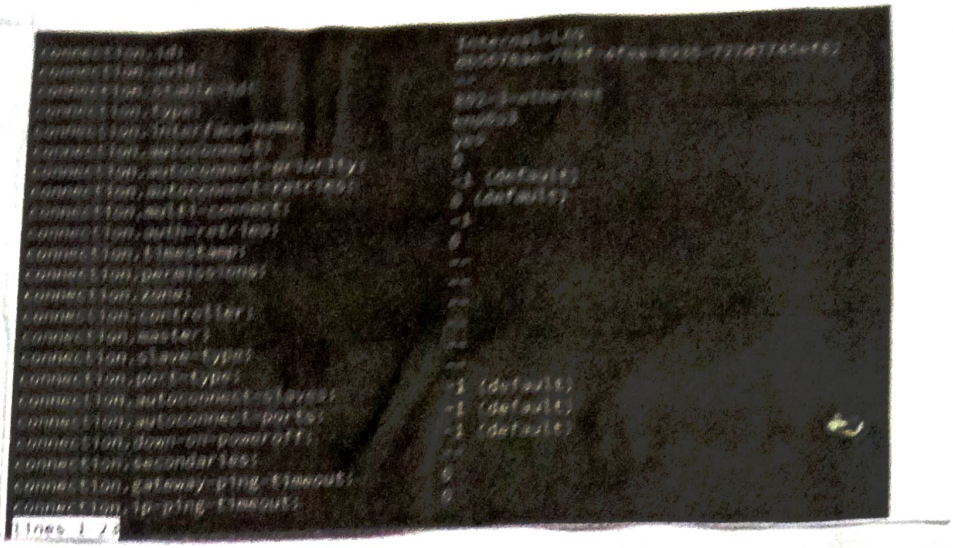
Activates the specified network Connection profile.

7. nmcli connection modify Internal LAN ipv6.method auto

Configures the connection to automatically obtain an IPV6 address using SLAAC.

8. nmcli connection modify Internal LAN ipv6.method manual ipv6.address 2001:db8::100/64 ipv6.gateway 2001:db8::1 ipv6.dns 2001:4860:4860::8888

Assigns a static IPV6 address, gateway, DNS server to the connection




```
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv4.method auto

ipv4.method:                            auto

[root@localhost ~]# nmcli connection modify  Internal-LAN ipv4.method manual ip
4.addresses 192.168.1.100/24 ipv4.gateway 192.168.1.1 ipv4.dns 8.8.8.8

ipv4.method:                            manual
ipv4.dns:                               8.8.8.8
ipv4.dns-search:                        --
ipv4.dns-options:                       --
ipv4.dns-priority:                      0
ipv4.addresses:                         192.168.1.100/24
ipv4.gateway:                           192.168.1.1

[root@localhost ~]# nmcli connection modify  Internal-LAN ipv6.method auto

ipv6.method:                            auto

[root@localhost ~]# nmcli connection modify  Internal-LAN ipv6.method manual ipv
6.addresses 2001:db8:1::100/64 ipv6.gateway 2001:db8:1::1 ipv6.dns 2001:4860:486
0::8888

ipv6.method:                            manual
ipv6.dns:                               2001:4860:4860::8888
ipv6.dns-search:                        --
ipv6.dns-options:                       --
ipv6.dns-priority:                      0
ipv6.addresses:                         2001:db8:1::100/64
ipv6.gateway:                           2001:db8:1::1
[root@localhost ~]# nmcli connection up Internal-LAN
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkMa
nager/ActiveConnection/3)
```